

Talking with the Satellites

Building a Low-Cost Ground Station with SatNOGS

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today



satellite rise

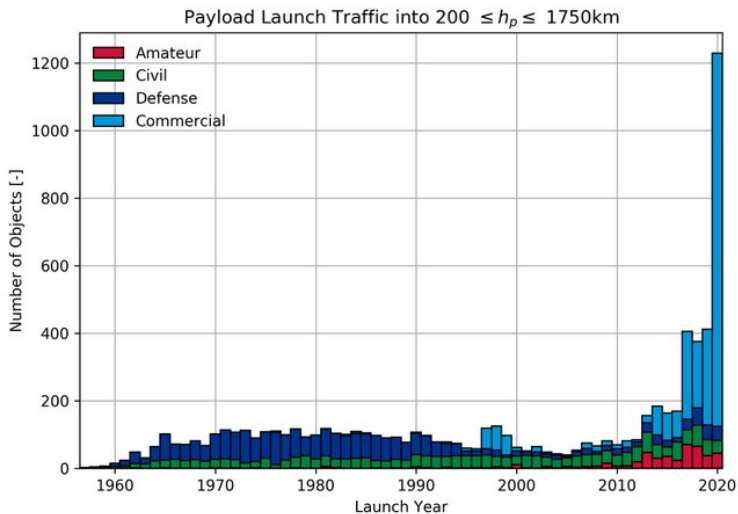


figure: number of satellites launched yearly into LEO (low earth orbit)



can i launch a satellite?

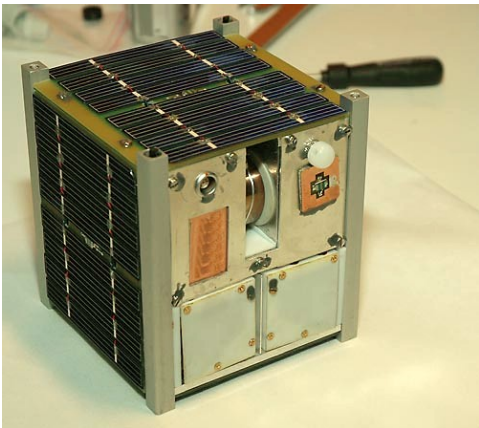


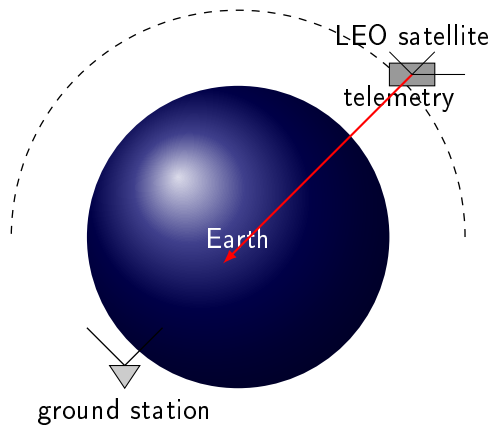
figure: a cubesat



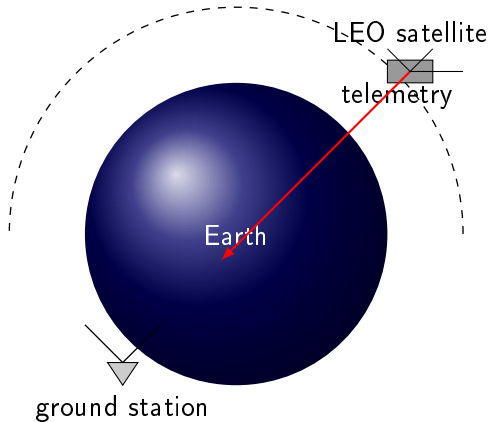
figure: a cubesat deployer



motivation



motivation

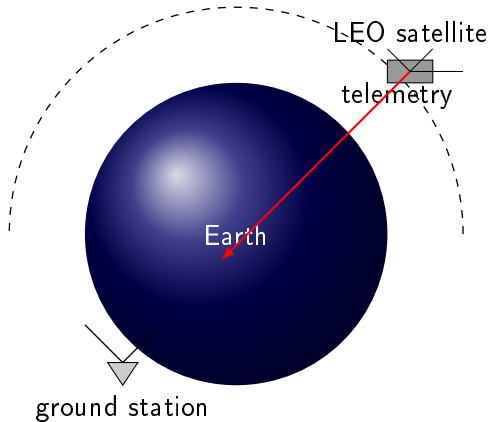


motivation:

- access real-time scientific data from space



motivation

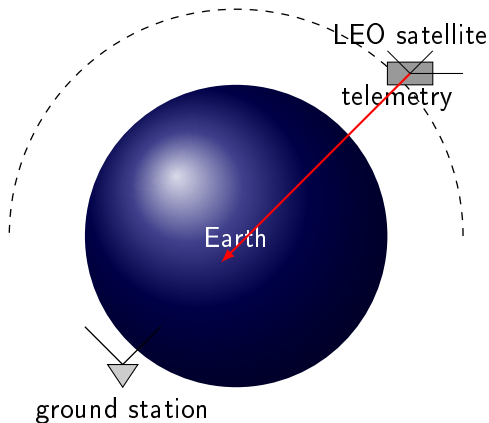


motivation:

- ▶ access real-time scientific data from space
- ▶ study RF propagation and satellite orbits



motivation

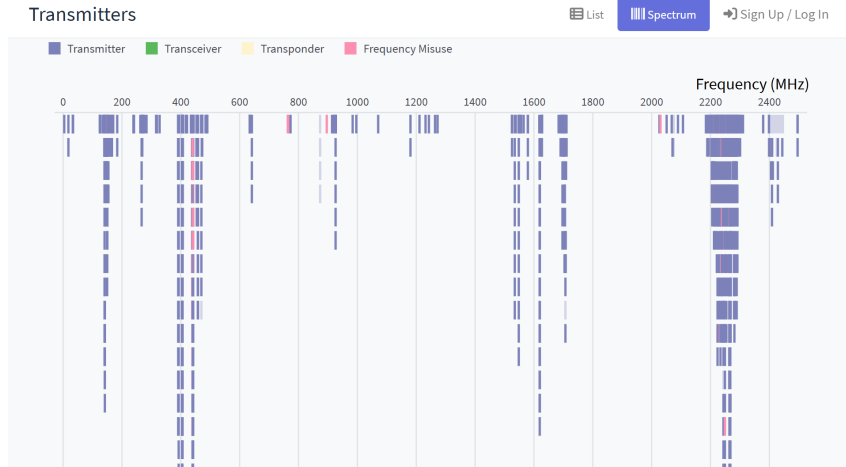


motivation:

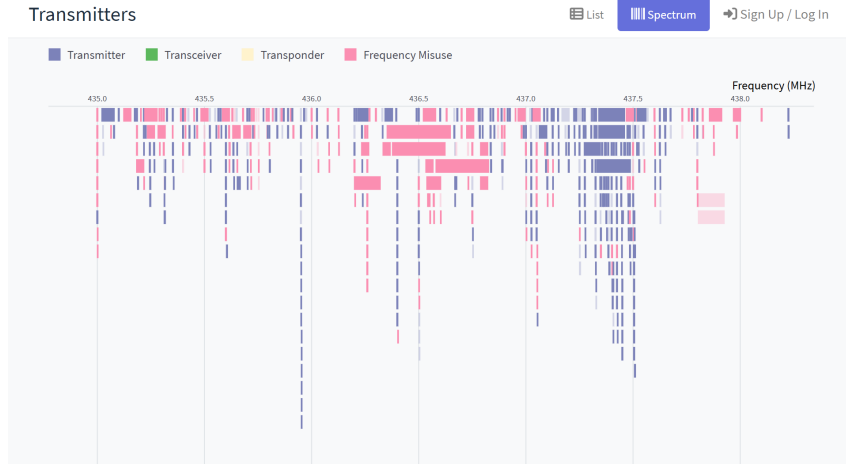
- ▶ access real-time scientific data from space
- ▶ study RF propagation and satellite orbits
- ▶ contribute to global open-source networks (SatNOGS)



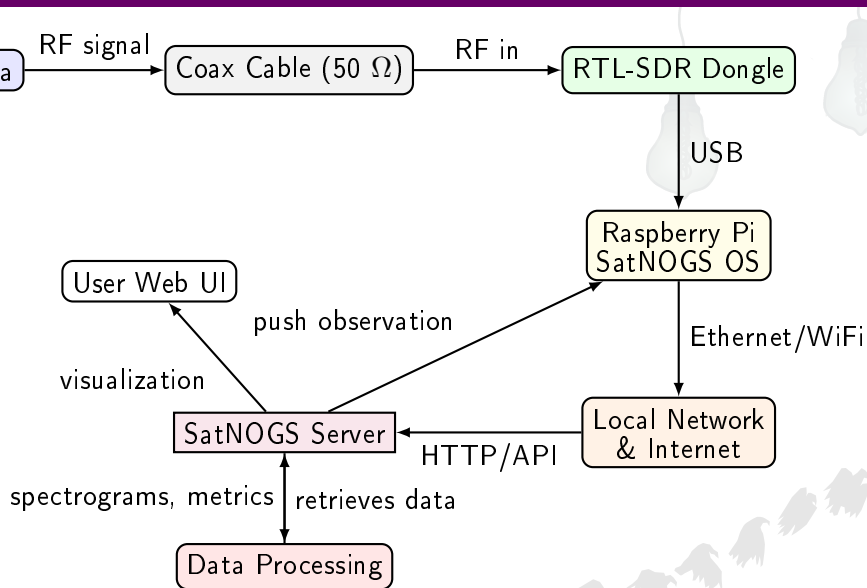
listening to waves



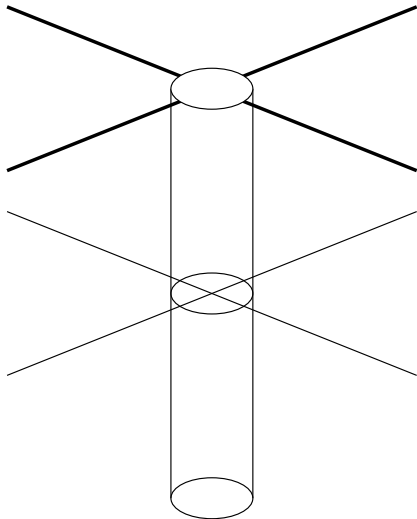
amateur frequencies



setup



antenna



- ▶ active (upper) rod length:
 $L_a = \frac{\lambda}{4} = \frac{c}{4f} = 16.3 \text{ cm}$
- ▶ passive (lower) rod length:
 $L_p = \frac{\lambda}{2} = \frac{c}{2f} = 34.6 \text{ cm}$
- ▶ distance between rods:
 $d = \frac{3\lambda}{8} = \frac{3c}{8f} = 25.8 \text{ cm}$
- ▶ interference: constructive



antenna

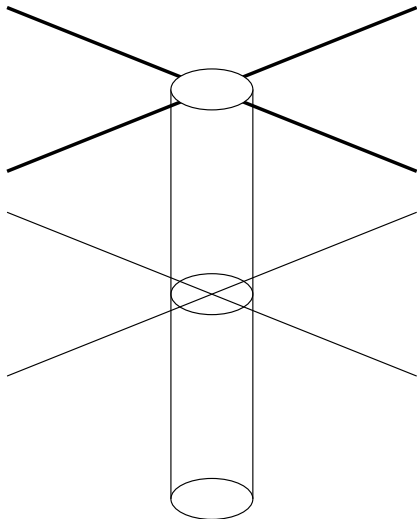


figure: the antenna with RTL-SDR dongle just outside the window



sdr + computer



figure: rtl-sdr dongle

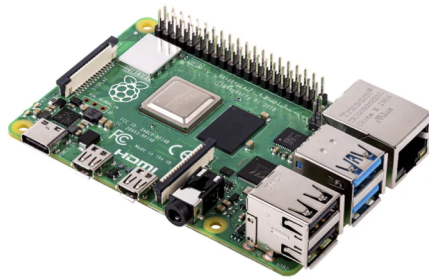


figure: raspberry pi



satnogs: concept

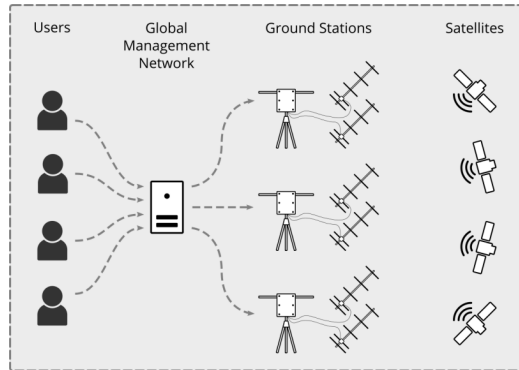
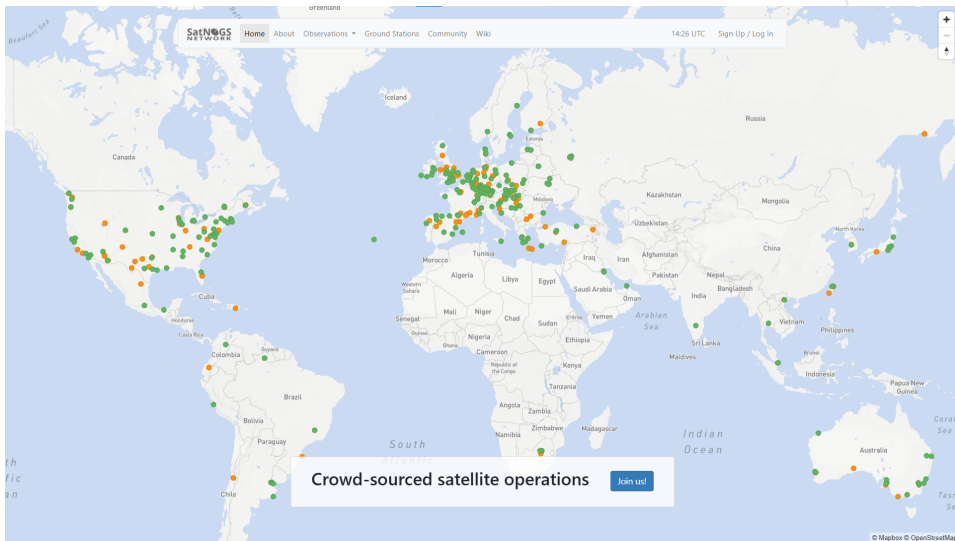


figure: overall view of satnogs (Satellite Network Open Ground Stations) concept



satnogs network



good observations I

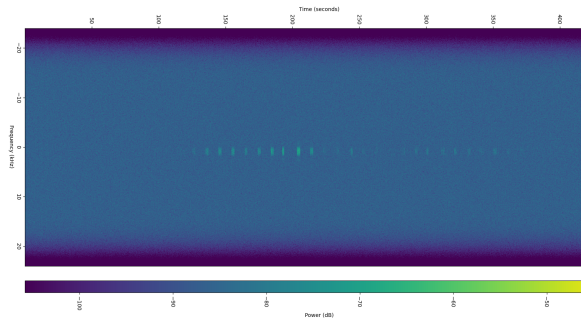


figure: satellite DhabiSat, transmitter: BPSK

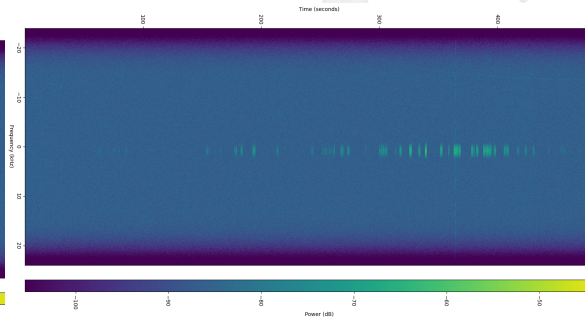


figure: satellite SITRO-AIS-9, transmitter: GMSK



good observations II

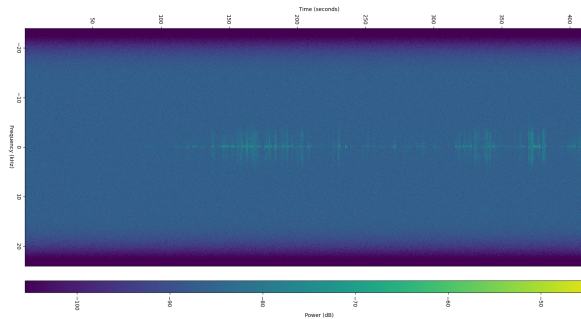


figure: satellite ISS, transmitter V/U FM

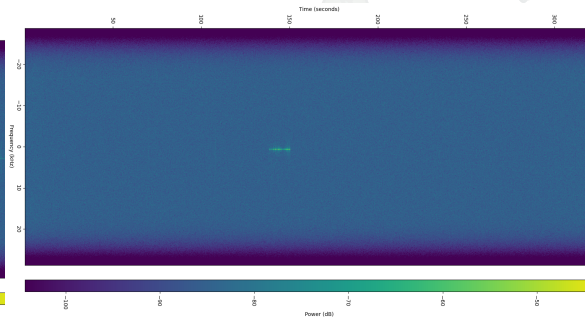


figure: satellite Reaktor Hello World,
transmitter: GFSK



decoding signal

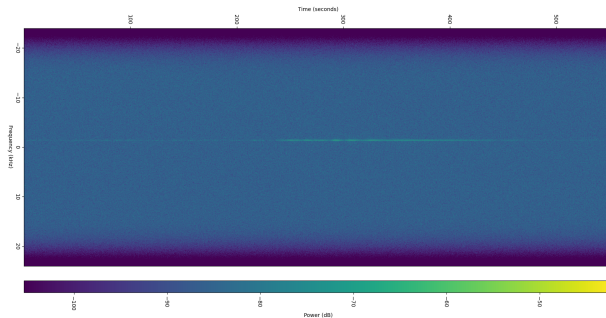


figure: satellite: LUSAT, transmitter: TLM

data_obs/2023/12/6/14/8662933/data_8662933_2023-12-06T14-33-49

EI I 5S S55

data_obs/2023/12/6/14/8662933/data_8662933_2023-12-06T14-35-28

5 SI E SEE I ESI IE5HI35 EE E S5HE IE



bad observations

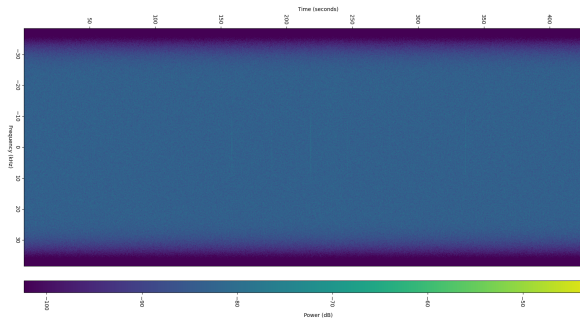


figure: bad result – no signal

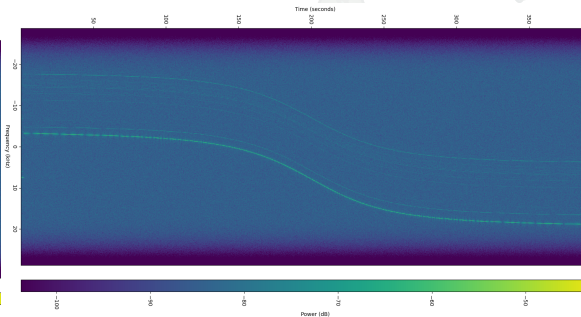


figure: also a bad result – tangent-like signal means terrestrial noise



G A R A .
G E . O F
C O M . .
. P L E X
. . I T Y



project co-funded by Garage of Complexity



bibliography

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- [1] Mateusz Winiarski. *ComplexityGarage/SatNOGS-station-*. Sept. 2023. URL: <https://github.com/ComplexityGarage/SatNOGS-station->.
 - [2] Musiał Alicja. *How to build a turnstile antenna for 435Mhz (and connect it to SatNOGS network) - tutorial for beginners* | *alicja.space*. en. Apr. 2021. URL: <https://alicja.space/blog/how-to-build-turnstile-antenna/>.
 - [3] Vitali Braun et al. "Fostering Collaborative Concepts in Space Debris Mitigation". In: Apr. 2021.
 - [4] Kevin Croissant et al. "An Updated Overview of the Satellite Networked Open Ground Stations (SatNOGS) Project". In: *Small Satellite Conference* (Aug. 2022). URL: <https://digitalcommons.usu.edu/smallsat/2022/all2022/172>.



thx for attention



<https://github.com/falcotinnunculus/przezroczka/blob/wladca/sempowisko2025/sempo.pdf>

